Title

Investigating advanced model-based reinforcement learning algorithms in the VizDoom environment.

Project Description

In our project we aim to test a new model-based reinforcement learning method on a complex environment such as the one offered by the VizDoom platform [1]. The main challenges proposed by the proposed environment are:

(i). High reward sparsity
(ii). High variability in the environment's possibilities (map, character and reward layout)
(iii). Partial knowledge due to purely perceptual information about the environment (first-person visual)

For environments of such complexity, the natural choice is to use world model-based approaches capable of autonomously learning a compact latent representation of the environments while maintaining a reasonable level of efficiency and, in most cases, outperforming traditional algorithms such as PPO [2]. However one of the main challenges that remains partially unsolved, even with world models, is efficient exploration in highly complex and sparse-reward environments, which requires the integration of specific solutions.

The objective of this project is to study how a latent world model can improve both the learning and the exploration in sparse-reward navigation tasks combining imagination-based policy with intrinsic rewards inspired by *curiosity-driven exploration.* [3] .

The curiosity modules implement sophisticated methods for exploration which, however, in addition to being computationally intensive, in several cases may find themselves "at odds" with what is "imagined" by the DREAMER model [4] . For this reason, we propose to adopt a recent variation of the DREAMERV3 [2] model known as DREAMERV3-XP [5] which is specifically designed for the exploration problem in sparse-reward environments.

The final objective of our work will therefore be to adapt the algorithm to the proposed environment and carry out a series of tests to ascertain the effectiveness of the new proposed approach compared to the basic version.

Group Components

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References

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